

# Acceptance Tests – WBOCRM

This document maps READ acceptance criteria to implemented test coverage.

All tests are located in backend/tests/ and results saved in backend/test\_results/.

## Test Suite Summary

| Suite      | File                                     | Tests                       | Status     |
|------------|------------------------------------------|-----------------------------|------------|
| Unit       | tests/unit/authService.test.js           | 6                           | ■ PASS     |
| Unit       | tests/unit/scoringService.test.js        | 6                           | ■ PASS     |
| Unit       | tests/unit/ticketService.test.js         | 6                           | ■ PASS     |
| Functional | tests/functional/api.test.js             | 6                           | ■ PASS     |
| BDD        | tests/features/auth_login.feature        | 4 scenarios                 | ■ PASS     |
| BDD        | tests/features/lead_registration.feature | 2 scenarios                 | ■ PASS     |
| BDD        | tests/features/lead_scoring.feature      | 3 scenarios                 | ■ PASS     |
| BDD        | tests/features/ticket_creation.feature   | 2 scenarios                 | ■ PASS     |
| BDD        | tests/features/duplicate_email.feature   | 2 scenarios                 | ■ PASS     |
| Total      |                                          | 35 tests / 13 BDD scenarios | ■ All pass |

## BDD Acceptance Scenarios

### **Feature: Lead Registration** (lead\_registration.feature)

#### **Scenario 1: Successfully register a new lead**

Given I am authenticated as a sales user  
When I submit a new lead with email "bdd\_lead@test.wbocrm" and name "BDD Lead"  
Then the response status should be 201  
And the response should contain a lead\_id

Covers: US-03, FR-LM-01

#### **Scenario 2: Reject duplicate email registration**

Given a lead with email "bdd\_dup@test.wbocrm" already exists  
When I submit a new lead with email "bdd\_dup@test.wbocrm" and name "Dup Lead"  
Then the response status should be 400  
And the response error should be "DUPLICATE\_EMAIL"

Covers: US-03, FR-LM-02

### **Feature: Support Ticket Creation** (ticket\_creation.feature)

#### **Scenario 1: Successfully create a support ticket**

Given I am authenticated as a support user  
And a lead exists in the system

When I create a ticket with description "BDD test ticket" and priority "Low"  
Then the response status should be 201  
And the response should contain a ticket\_id

Covers: US-11, FR-ST-01

#### **Scenario 2: Update a ticket status**

Given an existing support ticket  
When I update the ticket status to "In Progress"  
Then the response status should be 200  
And the ticket status should be "In Progress"

Covers: US-12, FR-ST-02

### ***Feature: User Authentication*** (auth\_login.feature)

#### **Scenario 1: Successful login with valid credentials**

When I log in with email "bdd\_auth\_sales@test.wbocrm" and password "BddAuth123!"  
Then the response status should be 200  
And the response should contain a token  
And the token payload should include role "sales"

Covers: US-01, FR-AUTH-01

#### **Scenario 2: Failed login with invalid password**

When I log in with email "bdd\_auth\_sales@test.wbocrm" and password "WrongPass999!"  
Then the response status should be 401

Covers: US-01, FR-AUTH-02

#### **Scenario 3: Rejected request without token**

When I request the leads list without a token  
Then the response status should be 401

Covers: US-01, FR-AUTH-03

#### **Scenario 4: Rejected request with insufficient role**

Given I am authenticated as a support user  
When I request the leads list  
Then the response status should be 403

Covers: US-02, FR-AUTH-04

### ***Feature: Lead Priority Scoring*** (lead\_scoring.feature)

#### **Scenario 1: High engagement metrics receive high score**

When I create a lead with metrics calls 20 meetings 5 budget "high" companySize "enterprise" emailOpens 20  
Then the response status should be 201  
And the lead priority\_score should be greater than 70

Covers: US-04, FR-SC-01

#### **Scenario 2: No engagement receives low score**

When I create a lead with metrics calls 0 meetings 0 budget "low" companySize "small" emailOpens 0  
Then the response status should be 201  
And the lead priority\_score should be less than 30

Covers: US-04, FR-SC-01

#### **Scenario 3: Leads list sorted by priority score descending**

When I retrieve all leads  
Then the response status should be 200  
And leads should be ordered by priority\_score descending

Covers: US-04, FR-LM-03, FR-SC-01

## Functional Test Coverage (`api.test.js`)

| Test                              | Expected                      | Covers       |
|-----------------------------------|-------------------------------|--------------|
| POST /api/leads – valid lead      | 201 + lead_id                 | US-03        |
| POST /api/leads – duplicate email | 400 DUPLICATE_EMAIL           | US-03        |
| GET /api/leads – sorted DESC      | 200 + ordered array           | US-04        |
| POST + PUT /api/tickets/:id       | 201 then 200 + status updated | US-11, US-12 |
| GET /api/leads – no token         | 401                           | US-01        |
| GET /api/leads – wrong role       | 403                           | US-02        |

## Unit Test Coverage

### *authService (6 tests)*

- `hashPassword` returns a bcrypt hash
- `comparePasswords` returns true for matching password
- `comparePasswords` returns false for wrong password
- `generateToken` returns a signed JWT
- `generateToken` payload contains `user_id` and `rbac_role`
- `verifyToken` returns the original payload

### *scoringService (6 tests)*

- Returns 0 for empty metrics
- Caps call score at 25 pts
- Caps meeting score at 25 pts
- Applies correct company size scores (small/medium/enterprise)
- Returns maximum 100 for fully maxed metrics
- Rounds to integer

### *ticketService (6 tests)*

- Creates a ticket successfully
- Returns `TICKET_NOT_FOUND` for missing ticket
- Returns `CONFLICT` when `updated_at` does not match
- Updates status successfully on valid `updated_at`
- Validates status enum values
- Assigns ticket to user correctly

## Non-Functional Requirements Verified

| NFR ID     | Requirement                          | Verification                                                             |
|------------|--------------------------------------|--------------------------------------------------------------------------|
| NFR-ST-05  | Optimistic locking on ticket updates | Unit test: <code>CONFLICT</code> on stale <code>updated_at</code>        |
| NFR-ST-12  | Dashboard auto-refresh every 2s      | <code>DashboardPage</code> <code>setInterval(2000)</code> implementation |
| NFR-SEC-01 | Passwords hashed with bcrypt         | Unit test: <code>hashPassword</code> returns non-plaintext               |
| NFR-SEC-02 | JWT auth on all protected routes     | Functional test: 401 without token                                       |
| NFR-SEC-03 | Role enforcement                     | Functional test: 403 for insufficient role                               |

|            |                              |                                                           |
|------------|------------------------------|-----------------------------------------------------------|
| NFR-ST-07  | GDPR/KVKK PII masking        | leadController masks email/name for support role          |
| NFR-ST-07  | Right-to-erasure             | DELETE /leads/:id/personal-data, /users/:id/personal-data |
| NFR-ST-15  | Ticket archiving (365+ days) | archiveService daily cron + manual POST /tickets/archive  |
| NFR-SEC-04 | Idle session timeout         | AuthContext 30-min inactivity auto-logout                 |
| NFR-ST-14  | Offline draft caching        | TicketPage localStorage pending ticket with retry         |

## Feature: Duplicate Email Detection (duplicate\_email.feature)

### Scenario 1: Reject lead with existing email

Given I am authenticated as a sales user  
And a lead with email "existing@test.wbocrm" already exists  
When I submit a new lead with email "existing@test.wbocrm"  
Then the response status should be 400  
And the response error should be "DUPLICATE\_EMAIL"

Covers: US-03, FR-LM-02

### Scenario 2: Frontend redirects to existing lead profile

Given I am on the leads page  
When I create a lead with a duplicate email  
Then I should see an error message containing "DUPLICATE\_EMAIL"  
And a link to the existing lead profile should be displayed

Covers: US-03, FR-LM-02

## Additional Coverage Notes

The following features are implemented and verified via code review but do not yet have dedicated BDD scenarios:

| Feature                 | Implementation                   | Verification Method           |
|-------------------------|----------------------------------|-------------------------------|
| Ticket archiving        | archiveService.js                | Code review + manual API test |
| Lead PII erasure        | leadController.erasePersonalData | Code review + manual API test |
| User PII erasure        | userController.erasePersonalData | Code review + manual API test |
| Idle session timeout    | AuthContext.jsx (30 min)         | Code review + manual UI test  |
| Offline draft caching   | TicketPage.jsx localStorage      | Code review + manual UI test  |
| PII masking for support | leadController.getAll/getById    | Code review + manual API test |
| Last admin guard        | userController.updateRole        | Code review + manual API test |